(i) $A \cup B$ $\Rightarrow$ at least one of the events A or B occurs

(ii) $A \cap B$ $\Rightarrow$ both events A and B occurs

(iii) $\overline{A} \cap \overline{B}$ $\Rightarrow$ Neither A nor B occur

(iv) $A \cap \overline{B}$ $\Rightarrow$ Event A occurs and B does not occur

(xii) Definition of Probability (Axiomatic approach)

Let E be an experiment. Let S be a sample space associated with E. With every event in S we associate a real number denoted by $P(A)$ called the probability of the event A satisfying the following axioms.

Axiom 1 : $P(A) \geq 0$

Axiom 2 : $P(S) = 1$

Axiom 3 : If $A_1, A_2, \supset, A_n$ be a sequence of n mutually exclusive events in S then

$$P(A_1 \cup A_2 \cup \ldots \cup A_n) = P(A_1) + P(A_2) + \ldots + P(A_n)$$

(xiii) Basic Theorems on probability

Theorem 1:

$P(\varnothing) = 0$ i.e., probability of an impossible event is zero.

Theorem 2:

Let S be the sample space and A be an event in S, then $P(\overline{A}) = 1 - P(A)$.

Theorem 3: Addition Theorem

If A and B are any two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

(xiv) Observation:

(i) If the two events A and B are mutually exclusive, then $A \cap B = \varnothing$

$\therefore P(A \cap B) = 0$

$\Rightarrow P(A \cup B) = P(A) + P(B)$

(ii) The addition theorem may be extended to any three events A, B, C and we have

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

It is believed that the students might be familiar with the above concepts and our present syllabus continues from the following.

8.3.2 Independent and Dependent events

(i) Independent Events

Two or more events are said to be **independent** when the outcome of one does not affect and is not affected by, the other. For example, if a coin is tossed twice, the result of the second throw would in no way be affected by the result of the first throw.

(ii) Dependent events

Are those in which the occurrence or non-occurrence of one event in any one trial affects the other events in other trials.

For example the probability of drawing a queen from a pack of 52 cards is $\frac{4}{52}$ or $\frac{1}{13}$. But if the card drawn (queen) is not replaced in the pack, the probability of drawing again a queen is $\frac{3}{51}$.

8.3.3 Conditional Probability

If two events A and B are dependent, then the conditional probability of B given that A as occurred already is

$$P(B/A) = \frac{P(A \cap B)}{P(A)}; \; P(A) \neq 0$$

Similarly $P(A/B) = \frac{P(A \cap B)}{P(B)}; \; P(B) \neq 0$

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